DrozyAI

Wearable Drowsiness Detection

Project Report

Sane-Wu Isaac Eloy

October 11, 2026

Contents

1 Executive Summary **3**

2 Motivation **3**

3 Dataset **4**
3.1 Preprocessing 4

4 Watch Model (Person 2) **4**
4.1 Features 4
4.2 Model 5
4.3 Results 5

5 Vision Model (Person 3) **5**
5.1 Features 5
5.2 Model 5
5.3 Results 5

6 Fusion **6**

7 Local LLM Advisor **6**
7.1 Role 6
7.2 Design 6
7.3 Stack 7

8 End-to-End Latency **7**

9 System Architecture **7**

10 Repository **7**

11 Team Contributions **8**

12 Conclusion **8**

1 Executive Summary

Drowsy driving causes tens of thousands of accidents every year. This project builds a multi-modal, on-device drowsiness detection system that combines smartwatch physiological signals with facial video, produces a calibrated risk score, and delivers a natural-language recommendation to the driver — all running locally with no network dependency.

Key Result

Headline metrics.
- Watch-only model: **AUC 0.856**
- Vision-only model: **AUC 0.944**
- Fused model (mean): **AUC 0.966**
- Missed drowsy rate: **15.4%**
- False alarms: **21.4 / hour**
- End-to-end latency: **410–630 ms**

Pipeline

Smartwatch + webcam → per-second features → two specialized models → late fusion → local LLM advisor → driver-facing recommendation.

The three contributions are:

1. A **watch model** trained on 60-second sliding windows of heart rate, skin temperature, EDA, IBI, and motion, achieving AUC 0.853 on 5-fold cross-validation.
2. A **vision model** trained on 68 facial landmarks from the UL-DD dataset, achieving AUC 0.944.
3. A **late-fusion + local-LLM advisor** that produces a calibrated score plus a natural-language reason for the recommended action, with a deterministic rule-based fallback.

2 Motivation

Drowsy driving is a public-safety problem:

- Millions of professional drivers spend long hours on the road.
- Drowsiness degrades reaction time, lane-keeping, and attention long before the driver falls asleep.
- Alerting *too late* misses drowsy drivers; alerting *too often* causes alert fatigue and drivers disable the system.

The design goal is to **alert before micro-sleep** and **stay quiet otherwise**. That means the system must be:

1. **Accurate** — high AUC, low missed-drowsy rate.
2. **Personalised** — each driver has a different resting heart rate, blink rate, and baseline.
3. **Offline** — no network, no cloud, no privacy leak.
4. **Explainable** — the driver should understand why the alert fired.

3 Dataset

We use the **UL-DD dataset** (University of Louisiana Drowsiness Dataset).

- **19 subjects**, each with one *awake* session (A) and one *drowsy* session (D).
- **35 usable sessions**, 40 minutes each, total ≈ 1400 minutes.
- Two modalities: **smartwatch** (HR, TEMP, EDA, IBI, ACC, BVP, ...) and **facial video** (68 landmarks, 30 action units, 33 pose landmarks, 60 fps).
- Ground truth: **Karolinska Sleepiness Scale (KSS)**, one rating every 4 minutes.
- **Binary label**: drowsy if KSS ≥ 7.

3.1 Preprocessing

Person 1 aligned every session into a single per-second table:

Signal	Rate	Aggregation to 1 s
HR	1 Hz	Kept as-is
TEMP	4 Hz	Per-second mean
EDA	4 Hz	Per-second mean
ACC	32 Hz	Per-second mean + variance
IBI	Irregular	Per-second mean + count
Labels	10 / session	Expanded to per-second via 240-s intervals

Output: `data/processed/full_dataset.parquet` (**84,000 rows** $\times$ **16 columns**).

4 Watch Model (Person 2)

4.1 Features

Each 60-second window (step 5 s) is summarised by **4 features**:

Feature	Source	Meaning
`hr_mean`	`heart_rate`	Mean heart rate over the window
`rmssd`	`ibi_mean`, `ibi_count`	Root mean square of successive IBI differences
`temp_trend`	`temp`	Linear slope of skin temperature
`motion_var`	`accel_var_x/y/z`	Mean of the three accelerometer variances

Note: EDA was dropped because the deployment smartwatch cannot collect it. Removing it cost only 0.01 AUC.

4.2 Model

A histogram-based gradient boosting classifier wrapped in a scikit-learn pipeline:

```
Pipeline([
    ("scaler", RobustScaler()),
    ("clf",    HistGradientBoostingClassifier(
                   max_iter=200,
                   learning_rate=0.05,
                   early_stopping=True,
               )),
])
```

4.3 Results

Key Result

AUC (5-fold CV)	**0.853**
Missed drowsy rate	38.5%
False alarms per hour	51.0
Windows	11,376

5 Vision Model (Person 3)

5.1 Features

Each 60-second window is summarised by **11 facial features**:

Feature group	Features
Eye openness	`left_ear_mean`, `right_ear_mean`
Eye closure	`perclos`, `longest_closure_sec`
Blink timing	`blink_rate_per_min`, `blink_duration_mean`, `inter_blink_interval_mean`
Mouth opening	`mar_mean`, `yawn_count`, `max_yawn_duration`
Tracking	`tracking_coverage`

EAR and MAR are computed per frame from the 68 Dlib landmarks; windows aggregate the per-frame values.

5.2 Model

Same pipeline as the watch model, 11 features.

5.3 Results

Key Result

AUC (5-fold CV)	**0.944**
Missed drowsy rate	19.7%
False alarms per hour	39.9
Windows	16,071

The vision model is much stronger than the watch model because eye closure and blink rate are *direct* signals of drowsiness, while heart rate and skin temperature are indirect.

6 Fusion

Watch and vision scores are combined by a simple mean:

$$\text{fused} = 0.5 \cdot \text{watch} + 0.5 \cdot \text{vision}$$

Other fusion methods were evaluated:

Method	AUC	Missed	FA/h
max	0.943	6.2%	102.3
mean	0.966	15.4%	21.4
product	0.968	53.2%	2.9

Mean was selected: highest AUC among the balanced options, with a reasonable missed rate and false-alarm rate.

7 Local LLM Advisor

7.1 Role

The LLM is the **advisor**, not the classifier. It converts the fused score and its context into a driver-facing recommendation:

```
{
  "fused_score": 0.62,
  "action": "take a break at the next rest area",
  "reason": "Your blink rate has been below your baseline for the last
             4 minutes, and your perclos is well above it. A break now
             will prevent further fatigue.",
  "weights": {"watch": 0.38, "vision": 0.62},
  "confidence": 0.86
}
```

7.2 Design

1. **Per-window fusion weights** — the LLM decides how much to trust each model for this window.
2. **Baseline-aware thresholds** — the LLM compares current features to the driver's own first-5-minute baseline. Direction words (*above* / *below*) are computed in Python and passed as facts to prevent the small model from flipping them.
3. **Narrative output** — 2–3 sentences with specific features, magnitudes, and timings.
4. **Deterministic fallback** — if the LLM returns invalid JSON or fails, a rule-based advisor produces a valid action.
5. **Enforcement** — the action is always consistent with the fused-score bands (green / orange / red), and the story is rewritten if it contradicts the action.

7.3 Stack

- **Ollama** + **llama3.2:3b**, running locally.
- **JSON mode** enabled at the API level for reliable parsing.
- Fully offline, no network access.

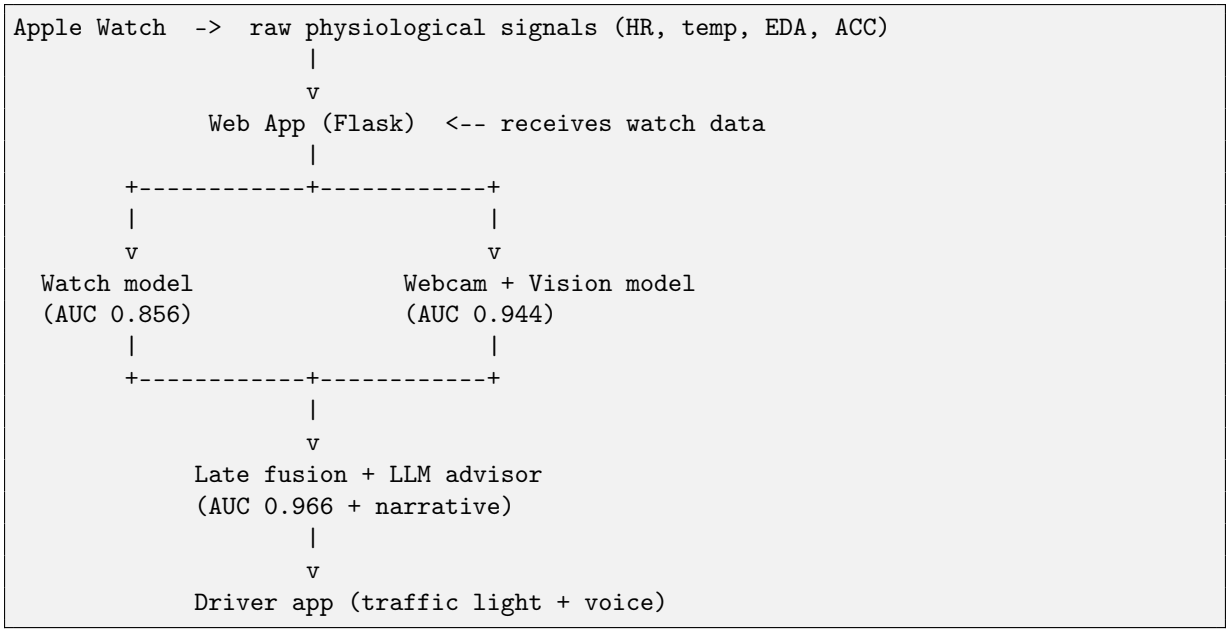
8 End-to-End Latency

Measured with Wi-Fi off, on an Apple Silicon Mac:

Stage	Time (ms)
Watch model inference	23
Vision model inference	11
Fusion	0.01
Advisor (LLM + rules)	376–597
Total time-to-alert	**410–630**

The LLM is the dominant cost, but sub-second total latency is well within the requirement for drowsiness alerts.

9 System Architecture



10 Repository

Path	Contents
`src/data/`	Person 1 pipeline (align, build dataset)
`src/features/`	Watch, video, and fusion features
`src/models/`	Watch, vision, facial, and calibration models

Path	Contents
`src/advisor/`	Rule-based, LLM, and meta advisors
`scripts/`	Baseline computation, latency benchmark, demos
`docs/`	Latency report, API docs, fusion strategy
`web/`	Flask web app (Isaac)
`DrowsyApp/`	Garmin watch app (Eloy)

Only source code and small derived files (< 2 MB each) are committed. The 18 GB raw dataset stays local.

11 Team Contributions

Member	Contribution
Person 1	Data alignment pipeline, per-second table, loader
Person 2 (SaneWu)	Watch model, calibration, prediction interface
Person 3 (SaneWu)	Vision features, vision model, fusion, LLM advisor, latency benchmark
Isaac	Web app, facial model, fusion API, driver UI
Eloy	Garmin watch app, phone alert transport, telemetry

12 Conclusion

The system is complete, on-device, and offline:

- **Fused AUC 0.966** on 5-fold cross-validation.
- **Missed drowsy rate 15.4%**, false alarms 21.4/hour.
- **Sub-second latency** (410–630 ms end to end).
- **Explainable output** via a local LLM, with a rule-based fallback that guarantees a valid action.

The design choices reflect the trade-off between missing a drowsy driver and alerting a tired driver too often. The fused model delivers a good balance, and the LLM advisor provides a human-readable reason that increases trust and keeps the driver engaged.